

On the Existence of Fundamental Ontic Positions: A Number-Theoretic No-Go Theorem

Alexander Thiele
Independent Researcher
`alexander.thiele@science-research-center.com`
ORCID: 0009-0006-6187-7019

DOI: 10.5281/zenodo.22092779

Abstract

We investigate a broad class of physical theories in which fundamentally localizable entities possess ontic positions that can be represented uniquely by numbers. This includes any framework in which spatial position is a fundamental physical attribute, independent of measurement, and can in principle be represented exactly by numerical coordinates.

Starting from a small set of physically motivated structural axioms, we formulate a measurement framework that distinguishes ontic positions from their numerical encoding. We show that every exact position measurement necessarily yields a finite numerical representation, while making no assumption that the underlying ontic position space is discrete.

The measurement framework provides a canonical arithmetic representation of all admissible exact position measurements with non-vanishing positional back-action. Once this representation has been established, the remainder of the proof proceeds entirely within elementary number theory. Under the stated assumptions, we derive a contradiction showing that an ontic position-based theory admitting such a unique numerical representation cannot consistently admit an exact position measurement with non-vanishing positional back-action. The key idea is that the set of admissible numerical values for a measurement is invariant under unit conversion. But this invariance will be violated once we do the math.

The contradiction is independent of dynamical laws, quantum-mechanical postulates, relativistic structure, probabilistic assumptions, and the particular realization of the measurement process. It therefore establishes a general number-theoretic no-go theorem for ontic position-based theories satisfying the stated assumptions and admitting at least one exact position measurement with non-vanishing positional back-action.

Keywords: ontic position; numerical representation; no-go theorem; measurement theory; quantum foundations; foundations of physics

1 Introduction

Physical theories routinely describe spatial positions by assigning real numbers to physical systems. This numerical representation is usually taken for granted and is therefore rarely examined as an independent structural assumption. In both classical and quantum physics,

positional quantities are commonly represented by real numbers, while the compatibility of this representation with exact physical positions is generally assumed rather than investigated.

The present work examines this assumption at a highly general level. Rather than considering a particular physical theory, we investigate the class of theories in which fundamentally localizable entities possess ontic positions that admit an injective numerical representation. No assumptions are made about dynamics, trajectories, probabilistic laws, quantum mechanics, relativistic spacetime, or specific measurement models beyond the structural framework introduced below.

The present result differs fundamentally from established no-go theorems on particle localization, such as Malament’s theorem and its subsequent extensions by Halvorson and Clifton, which constrain localizable particles within relativistic quantum theory under assumptions concerning relativistic covariance, energy, localizability, and locality [1, 2].

In contrast, the present argument does not rely on relativistic or quantum-mechanical structure. It addresses the more general question of whether fundamental ontic positions can consistently possess the numerical representation structure introduced below.

An injective numerical representation assigns a unique numerical value to each ontic position and thereby provides an unambiguous numerical description of the underlying position space. It is important to distinguish this numerical representation from the ontic positions themselves. The existence of physical positions is a statement about the physical world, whereas their numerical representation is an additional mathematical structure used to describe that world. The central question of this work is whether these two structures can consistently coexist under a minimal set of physically motivated assumptions.

The question addressed here is therefore not whether physical systems can be localized operationally, but whether fundamentally localizable entities can consistently be assigned unique numerical position values under the below stated axioms and the measurement framework introduced below.

To address this question, we first introduce an axiomatic framework for ontic position-based theories. We then develop a representation-independent measurement framework that separates ontic positions from their numerical encoding. This framework shows that exact position measurements necessarily yield finitely representable numerical results, while making no assumption that the underlying ontic position space itself is discrete.

From these structural properties, we derive a canonical arithmetic representation of exact position measurements with non-vanishing positional back-action. Moreover, we are going to prove that the admissible set of certain numerical measurement-specific values must be invariant under unit conversion. Once this measurement framework has been established, the remainder of the proof proceeds entirely within elementary number theory. The resulting contradiction, which is the violation of the invariance under unit conversion, therefore follows solely from the arithmetic structure implied by the preceding physical assumptions.

The resulting theorem establishes a general no-go result for ontic position-based theories satisfying the proposed axioms and admitting at least one exact position measurement with non-vanishing positional back-action. Its consequences are independent of quantum mechanics, relativistic structure, dynamical laws, probabilistic assumptions, and specific measurement interactions.

Although the theorem is formulated for injectively representable ontic positions, many physical theories naturally describe spatial position using coordinate-based numerical representations. Consequently, the class of ontologies not ruled out by the theorem may be smaller

than the formal statement alone suggests.

The contradiction derived below does not depend on any particular physical theory but follows from the structural assumptions of the framework introduced in this work.

2 Axioms, Hypothesis and Basic Definitions

Definition 2.1 (localizable entity). A physical entity is called fundamentally localizable if, at every instant of time, it possesses a unique ontic spatial position that is independent of observation, measurement, or epistemic access. The existence of such a position constitutes an objective property of the physical system and does not depend on any measurement procedure or numerical representation.

Hypothesis 2.2 (ontic position). The present work investigates physical theories satisfying the following ontic position hypothesis:

Every fundamentally localizable entity possesses a unique ontic position at every instant of time. This ontic position exists independently of measurement and admits an injective numerical representation.

The following axioms formalize this hypothesis.

Axiom 2.3 (Existence). Every fundamentally localizable physical entity possesses an ontic position at every instant of time.

Axiom 2.4 (Uniqueness). At every instant of time, each fundamentally localizable entity possesses exactly one ontic position.

Axiom 2.5 (Exact measurability). In an idealized localization limit, where neither fundamental nor technical measurement limitations (e.g. uncertainty relations, detector resolution, or noise) are present, every ontic position is, in principle, exactly measurable.

This idealized limit serves solely as a structural reference model. No assumptions are made concerning the practical realization of such measurements.

The role of this idealized assumption is discussed further in Appendix B. There it is shown that reintroducing measurement uncertainties, including those associated with the Heisenberg uncertainty principle, does not remove the contradiction. The idealized limit is considered because it provides the strongest case for the existence of fundamental numerical positions. The no-go result therefore does not depend on exact measurability.

Axiom 2.6 (Injective representation). There exists a set X of ontic positions and an injective numerical representation

$$n : X \rightarrow \mathbb{R},$$

assigning a unique real number to every ontic position.

Injectivity ensures that distinct ontic positions always possess distinct numerical representations.

No assumptions are made regarding dynamics, probability, trajectories, or temporal evolution.

Unless explicitly stated otherwise, the analysis is restricted to one spatial dimension. The generalization to $\mathbb{R}^k$ is immediate since the proof is component-wise and does not rely on dimension-specific properties.

Definition 2.7 (Ontic Position System). A physical system is called an ontic position system if it satisfies Axioms 2.3–2.6.

Accordingly, every fundamentally localizable entity possesses a unique ontic position at every instant of time. In the idealized localization limit introduced in Axiom 2.5, this position is, in principle, exactly measurable.

Definition 2.8 (Ontic Position-Based Theory). A physical theory is called an ontic position-based theory if there exist

$$P : S \rightarrow X$$

and

$$n : X \rightarrow \mathbb{R},$$

where

- S denotes the state space,
- X denotes the ontic position space,
- P assigns an ontic position to every physical state,
- n is an injective numerical representation assigning a unique real number to every ontic position.

The mapping

$$P : S \rightarrow X$$

assigns ontic positions to physical states.

The mapping

$$n : X \rightarrow \mathbb{R}$$

assigns unique numerical representations to ontic positions.

This distinction between physical positions and their numerical representation is fundamental throughout the remainder of this work.

No assumptions are made concerning dynamics, trajectories, probabilistic structure, or temporal evolution.

3 Representation Framework for Exact Position Measurements

Having established the structural framework of ontic position-based theories, we now turn to the numerical representation of ontic positions in the context of physical measurements.

While the preceding axioms postulate the existence of ontic positions as objective properties of physical systems, measurements do not provide direct access to these ontic positions themselves. Instead, every measurement produces a numerical representation associated with a positional attribute.

The purpose of this section is therefore not to introduce additional physical assumptions but to formalize the numerical structure of exact position measurements. In particular, we distinguish between ontic positions, primary measurement results, and their subsequent mathematical interpretation. This distinction allows the physical assumptions above to be expressed mathematically.

We shall show that every exact positional measurement necessarily admits a finite numerical representation. Importantly, this restriction concerns only the numerical description produced by the measurement process. It neither implies any discreteness of the underlying ontic position space nor does it introduce any additional assumption concerning ontic localization beyond the axioms already introduced.

Once this representation framework has been established, all subsequent arguments leading to the no-go theorem become purely number-theoretic.

Definition 3.1 (Measurement). A measurement is understood as an operational procedure that yields a numerical representation of a positional attribute.

Throughout this work, numerical representations are expressed in positional decimal notation. Accordingly, a finite numerical representation means a finite positional decimal representation.

Under the assumptions introduced below, we show that this numerical representation is necessarily finite.

This finite representation does not imply that ontic position space is discrete. It is simply the numerical representation produced by the measurement process.

Accordingly, measurements do not directly yield ontic positions themselves, nor arbitrary elements of $\mathbb{R}$. Instead, they yield finite numerical representations that can subsequently be interpreted within an appropriate mathematical model as representations of ontic positions.

Definition 3.2 (Primary Measurement Result). A primary measurement result is understood as the immediate outcome of a measurement process before any further mathematical operations, model assumptions, or theoretical interpretations are applied.

A primary measurement result is therefore the information that is directly available to an observer after completion of the measurement and can be physically stored, displayed, or communicated.

Position and length measurements belong to the class of primary measurements. For example, if the position of a localized physical entity is reported as a numerical value, this value itself is the primary measurement result.

By contrast, quantities such as velocity, acceleration, density, electric power, and other derived observables generally result from mathematical processing of one or more primary measurement results and are therefore secondary quantities.

Throughout the remainder of this work, the term measurement result always refers to a primary measurement result unless explicitly stated otherwise.

Lemma 3.3. *Every primary measurement result necessarily possesses a finite description length.*

In particular, no primary measurement result can consist of an infinite decimal expansion.

Proof. Let R denote the primary result of an arbitrary measurement process.

By Definition 3.2, R is the information that becomes directly available to an observer immediately after completion of the measurement. Consequently, R must possess a physical representation, for example as the state of a memory device, a detector output, a display, a written record, or a communication signal.

Every completed physical representation necessarily consists of a finite sequence of symbols. Otherwise, the complete measurement result could neither be stored, completely displayed, nor completely transmitted to an observer.

Assume, for contradiction, that the complete decimal expansion of R itself forms part of the primary measurement result and contains infinitely many digits. Then the physical representation of R would necessarily require an infinite number of symbols. Such a representation could never exist as a completed physical record and therefore could never become completely available to an observer.

This contradicts the definition of a primary measurement result.

The conclusion is independent of the technical performance of measuring devices. Even an idealized measuring device with infinite spatial resolution could at most distinguish arbitrarily small positional differences. It would still produce a primary measurement result only after the acquired information has been physically represented. Since every physical representation necessarily contains finitely many symbols, infinite spatial resolution does not imply an infinitely long measurement result.

Hence, no primary measurement result can contain an infinite decimal expansion as part of its physical representation.

Therefore, every primary measurement result necessarily possesses a finite description length. $\square$

3.1 Representation Independence of Measurement Resolution

The preceding lemma does not rely on any technical limitation of present-day measuring instruments.

One might argue that an idealized measuring device possessing infinite spatial resolution could determine positional values with arbitrarily high precision. Even if such an idealization is admitted, the conclusion of Lemma 3.3 remains unchanged.

Infinite spatial resolution merely allows arbitrarily small positional differences to be distinguished. It does not imply that the resulting measurement output itself becomes available as an infinite decimal expansion.

For a measurement result to exist, the acquired information must become physically accessible to an observer. This necessarily requires a finite physical representation. An infinite decimal expansion would require an infinite number of symbols and therefore could neither be completely stored, completely displayed, nor completely communicated.

Consequently, the restriction established in Lemma 3.3 follows from the concept of a completed measurement result itself and not from the finite resolution of measuring devices.

Even an idealized measuring device with infinite spatial resolution can therefore produce only finitely encoded primary measurement results.

3.2 Coordinate Freedom

The numerical value assigned to a measured position depends on the choice of coordinate system used to represent the underlying ontic position.

The choice of coordinate origin and length unit constitutes a purely representational convention and possesses no physical significance. It neither alters the ontic position itself nor affects the outcome of the physical measurement process.

Throughout this work, the coordinate origin and length unit may therefore be chosen freely whenever this simplifies the numerical representation of exact measurement results.

Proposition 3.4 (Coordinate Freedom). *For every fundamentally localizable entity, one can always choose a coordinate origin such that the corresponding exact position measurement possesses a finite numerical representation.*

Proof. Let x denote the ontic position of a fundamentally localizable entity. Since there exists a finitely representable number f , there exists an affine coordinate transformation

$$x' = \kappa x + \lambda, \quad \kappa \neq 0.$$

Indeed, choosing

$$\lambda = f - \kappa x,$$

yields

$$x' = f.$$

Hence, every fixed ontic position admits an exact finite numerical representation in a suitably chosen coordinate system.

Consequently, the occurrence of an infinite decimal expansion is not an invariant property of the physical position itself but solely of the chosen numerical representation. Therefore, every independently localizable entity admits an exact finite numerical representation as a primary measurement result. $\square$

Example 3.5 (Choice of Length Unit). Consider two idealized measuring devices observing the same localizable entity from a common coordinate origin.

Suppose the first device employs the length unit A , whereas the second device uses the unit B , related by

$$3A = 1B.$$

Assume that the entity is located exactly at the position $1A$. The first measuring device therefore produces the exact primary measurement result $1A$. The second measuring device observes the identical physical position but must represent it numerically as

$$\frac{1}{3}B.$$

Since $0.333\dots$ has no finite decimal representation, two possibilities remain: Either the measurement process terminates after finitely many digits, yielding only an approximation of the exact position, or the measurement process never produces a completed numerical result. Neither possibility provides an exact primary measurement result in the chosen unit.

This example illustrates that the finiteness of a numerical representation depends on the chosen unit, whereas the underlying ontic position remains unchanged.

Accordingly, throughout the subsequent analysis, exact primary measurement results are represented in units chosen such that their numerical representations are finite.

Approximate measurement results are excluded from the subsequent analysis because the no-go theorem concerns only exact positional assignments.

3.3 Measurement Back-Action

Operationally, a measurement involves a physical interaction through which information about the system is obtained. The present analysis considers exact position measurements for which the measurement interaction induces a non-vanishing change in the ontic position,

$$x_{\text{ba}} := \bar{x} - x \neq 0,$$

which we refer to as the measurement back-action.

No claim is made that every conceivable exact position measurement must necessarily satisfy this condition. Measurements with $x_{\text{ba}} = 0$ are not covered by the arithmetic construction developed below, since in that case the three-term measurement equation degenerates to $\bar{x} = x$.

Importantly, the existence of individual measurements with $x_{\text{ba}} = 0$ does not provide an escape from the no-go result. The argument requires only that at least one admissible exact position measurement with $x_{\text{ba}} \neq 0$ exists. Consequently, avoiding the present no-go result through vanishing positional back-action would require

$$x_{\text{ba}}^{(M)} = 0$$

for every admissible exact position measurement M .

No assumptions are made regarding the physical origin, magnitude, or dynamical mechanism of the measurement back-action.

Let x denote the initial ontic position immediately before the measurement interaction and $\bar{x}$ the final ontic position immediately after completion of the measurement. The measurement back-action is denoted by x_{ba} . It represents the positional change induced by the measurement interaction itself. Consequently, every exact position measurement considered in the present analysis satisfies the fundamental relation

$$\bar{x} = x + x_{\text{ba}}. \tag{3.1}$$

where

- x denotes the initial ontic position,
- $\bar{x}$ denotes the final ontic position,
- $x_{\text{ba}} \neq 0$ denotes the measurement back-action.

Equation (3.1) constitutes the fundamental measurement equation underlying the remainder of this work.

3.4 Exact Numerical Representation of the Fundamental Measurement Equation

According to Lemma 3.3, every primary measurement result possesses a finite numerical representation.

The initial position x therefore admits a finite numerical representation.

The same conclusion holds for the remaining two quantities appearing in Eq. (3.1).

To see this, consider two exact position measurements performed consecutively on the same localizable entity.

For the first measurement,

$$\bar{x}_1 = x_1 + x_{\text{ba},1}.$$

Immediately afterwards, a second exact position measurement yields

$$\bar{x}_2 = x_2 + x_{\text{ba},2}.$$

Since the second measurement starts precisely where the first measurement ends,

$$x_2 = \bar{x}_1.$$

By Lemma 3.3, the primary measurement result x_2 possesses a finite numerical representation.

Hence the final position $\bar{x}_1$ must also possess a finite numerical representation.

Since this argument is independent of the particular measurement under consideration, every final ontic position appearing in Eq. (3.1) necessarily admits a finite numerical representation.

Because both $\bar{x}$ and x possess finite numerical representations, the measurement back-action must likewise admit a finite numerical representation by Eq. (3.1).

The preceding observations immediately imply the following lemma.

Lemma 3.6. *When expressed in the unit associated with the corresponding primary measurement result, every exact position measurement satisfies a representation of Eq. (3.1) in which*

- *the initial ontic position,*
- *the final ontic position,*
- *and the measurement back-action*

all possess finite numerical representations.

Proof. The initial position is finite by Lemma 3.3.

The finiteness of the final position follows from the argument based on two consecutive exact measurements established above.

The measurement back-action is obtained as the difference of two finitely represented quantities according to Eq. (3.1).

Therefore, all three quantities admit finite numerical representations. $\square$

Definition 3.7 (Physical Measurement Equation). An exact position measurement represented in the unit associated with its corresponding primary measurement result is called a *physical measurement equation* if all three quantities appearing in Eq. (3.1),

$$\bar{x}, x, x_{\text{ba}},$$

possess finite numerical representations.

Only under this condition can all numerical values occurring in Eq. (3.1) arise directly as exact primary measurement results.

A mathematically equivalent representation obtained, for example, by a change of length unit may contain one or more quantities with infinite decimal expansions.

Such representations are referred to as *unphysical measurement equations*, since at least one of the quantities $\bar{x}$, x , or x_{ba} is represented by a numerical value that can no longer occur as an exact primary measurement result but arise only through subsequent mathematical transformations.

3.5 Arithmetic Representation of Physical Measurement Equations

From this point onward, we consider only physical measurement equations in the sense of Definition 3.7.

By Lemma 3.6, each quantity appearing in the fundamental measurement equation (3.1) possesses a finite numerical representation. Consequently, every term belongs to the class of finitely representable rational numbers. The remainder of the proof therefore concerns exclusively the arithmetic structure of such representations.

The decimal system is used throughout this work solely as an illustrative example. All subsequent arguments remain valid in any positional numeral system after replacing the corresponding prime factors of the base.

In the decimal system, every finitely representable number possesses a unique reduced fractional representation whose denominator contains no prime factors other than 2 and 5.

Accordingly, the three quantities appearing in Eq. (3.1) necessarily admit representations of the form

$$\bar{x} = \frac{a}{2^i 5^j}, \quad (3.2)$$

$$x = \frac{c}{2^k 5^l}, \quad (3.3)$$

$$x_{\text{ba}} = \frac{e}{2^m 5^n}, \quad (3.4)$$

where

$$a, c, e \in \mathbb{Z} \setminus \{0\},$$

$$i, j, k, l, m, n \in \mathbb{Z},$$

and

$$\gcd(a, 2^i 5^j) = \gcd(c, 2^k 5^l) = \gcd(e, 2^m 5^n) = 1.$$

Negative exponents are permitted and represent factors of 2 or 5 in the numerical value that are not contained in the numerators a , c , and e . Thus, the above expressions describe all finitely representable rational numbers.

Substituting Eqs. (3.2)–(3.4) into the fundamental measurement equation, Eq. (3.1), yields

$$\frac{a}{2^i 5^j} = \frac{c}{2^k 5^l} + \frac{e}{2^m 5^n}. \quad (3.5)$$

Equation (3.5) constitutes the canonical arithmetic representation of every exact physical position measurement.

From this point onward, the proof no longer depends on the physical interpretation of the three quantities but only on the arithmetic properties of Eq. (3.5).

All subsequent arguments used in the proof of the no-go theorem are therefore purely number-theoretic.

3.6 Choice of Coordinate Origin

The numerical values appearing in Eq. (3.5) depend on the arbitrary choice of coordinate origin.

Since the coordinate origin possesses no physical significance, it may be chosen freely to eliminate unnecessary arithmetic degeneracies.

Throughout the remainder of this work, the coordinate origin is therefore selected according to the following conditions.

Condition 3.8. The coordinate origin is chosen such that

$$x \neq 0, \quad \bar{x} \neq 0.$$

This excludes trivial two-term representations of Eq. (3.5).

Since the position of the coordinate origin is entirely conventional, this restriction has no physical consequences.

Condition 3.9. The coordinate origin is furthermore chosen such that

$$|a| \neq |c|, \quad |a| \neq |e|, \quad |c| \neq |e|.$$

If one of these equalities occurs, the coordinate origin can always be translated by a suitably chosen finite displacement. Such a translation changes only the numerical representation of the measurement while leaving the underlying physical measurement unchanged. Consequently, these equalities may always be excluded without loss of generality.

Note that a coordinate displacement adds the same numerical value to both $\bar{x}$ and x . The measurement back-action x_{ba} , being the fixed positional difference between $\bar{x}$ and x , therefore remains unchanged, since $x_{\text{ba}} = \bar{x} - x$.

Lemma 3.10. Equation (3.5) may always be assumed to be in maximally reduced form, i.e.,

$$\gcd(a, c) = 1, \quad \gcd(a, e) = 1, \quad \gcd(c, e) = 1.$$

Proof. If all three numerators share a common factor distinct from the prime factors 2 and 5, this factor may be cancelled simultaneously from all three fractions without affecting Eq. (3.5).

Suppose, instead, that only two of the three numerators in Eq. (3.5) possess a common factor

$$y \neq 2^r 5^s.$$

Consequently, the third numerator is coprime to y .

Cancelling this factor from only two fractions necessarily introduces the factor y into the denominator of the remaining fraction. Since y contains a prime factor different from 2 or 5,

the remaining fraction would no longer possess a finite decimal representation. This contradicts Lemma 3.6.

Hence Eq. (3.5) may always be assumed to satisfy

$$\gcd(a, c) = \gcd(a, e) = \gcd(c, e) = 1.$$

□

Equation (3.5), together with Lemma 3.10 and Conditions 3.8–3.9, completely specifies the arithmetic structure of every exact physical position measurement considered in this work. All subsequent results follow exclusively from elementary properties of this arithmetic structure.

3.7 Representation Equivalence under Unit Transformations

The arithmetic structure derived above is independent of the particular unit of length chosen for its numerical representation.

Mathematically, a unit conversion corresponds to a linear scaling

$$x_{\text{conv}} = b x_{\text{orig}},$$

where

$$b$$

denotes the conversion factor.

A purely mathematical unit conversion imposes no restriction on the resulting numerical values.

Physical measurements, however, require additional consistency conditions.

Whether a converted representation remains physically admissible depends on the numerical representation of the converted quantities rather than on the scaling operation itself.

Finite representability is not assumed to be an intrinsic property of a physical position independently of the chosen reference unit. As shown by Proposition 3.4, the numerical representation of a physical position depends on the chosen coordinate representation and may be finite in one representation and non-terminating in another.

The physical significance of finite representability instead follows from Lemma 3.3. An exact primary measurement result must possess a finite description length in the unit in which it is physically represented. Accordingly, a converted equation whose quantities are not all finitely representable remains mathematically valid, but it cannot be represented by an exact physical measurement equation in that unit. Thus, the numerical values for $\bar{x}$, x , and x_{ba} in a converted measurement equation can only be reproduced by a measurement device when those values are finitely representable.

The distinction is therefore not between physically admissible and inadmissible units. It is between a purely mathematical representation obtained by unit conversion and a physical measurement equation whose numerical values can occur as exact primary measurement results.

Definition 3.11 (Original and Converted Measurement Equations). A physical measurement equation obtained directly from an exact measurement is called the *original measurement equation*.

Since every primary measurement result possesses a finite numerical representation, every original measurement equation is necessarily physical.

A converted measurement equation is obtained from the original measurement equation by a unit transformation.

The converted equation is called

- *physical*, if all three quantities

$$\bar{x}, x, x_{\text{ba}}$$

remain finitely representable,

- *unphysical*, if at least one of the three quantities no longer possesses a finite numerical representation and therefore cannot occur as an exact primary measurement result.

Lemma 3.12. *Every finitely representable numerical value can occur independently as the numerical representation of the final position $\bar{x}$, the initial position x , and the measurement back-action x_{ba} .*

Consequently, the admissible numerical representations of all three quantities are given by the common set

$$D = \left\{ \frac{p}{2g5^h} \mid p \in \mathbb{Z} \setminus \{0\}, g, h \in \mathbb{Z} \right\},$$

and therefore

$$D_{\bar{x}} = D_x = D_{x_{\text{ba}}} = D.$$

Proof. By the preceding measurement framework and the idealized localization limit established in the previous section, all restrictions originating from finite detector resolution and uncertainty-related limitations are removed. Consequently, finite representability itself is the only remaining structural restriction on admissible numerical values.

The attainable numerical representations therefore depend only on the chosen physical configuration of the measurement process. Its initial conditions, boundary conditions, detector realization, and interaction mechanism may be chosen freely.

Since the measurement back-action depends on the complete experimental setup, the numerical values of all three quantities can be chosen independently by changing the measurement configuration.

Every finitely representable numerical value can therefore occur independently as the numerical representation of each of the three quantities.

It follows immediately that all three quantities possess the same admissible numerical set,

$$D_{\bar{x}} = D_x = D_{x_{\text{ba}}} = D.$$

□

Lemma 3.13. *For every individual measurement M , the admissible numerical set is invariant under a change of the reference length unit.*

Thus, for an original measurement equation and its corresponding physical converted measurement equation,

$$D_{\text{orig}}^{(M)} = D_{\text{conv}}^{(M)} = D.$$

Proof. Consider an individual measurement M performed with a measuring device calibrated in a reference length unit A .

By Lemma 3.12, every finitely representable numerical value can occur independently as the numerical representation of $\bar{x}$, x , and x_{ba} .

This result applies to the admissible numerical values of an individual measurement. The physical configuration of the measurement process may be chosen freely by changing its initial conditions, boundary conditions, detector realization, or interaction mechanism. Consequently, for an original measurement equation, every numerical value in D can be obtained by an appropriate choice of the measurement configuration.

Therefore,

$$D_{\text{orig}}^{(M)} = D.$$

Now consider the reference length unit A . The reference unit A need not be an original physical standard. It may itself have been obtained by converting an earlier reference unit A' during the manufacturing or calibration of the device.

A measurement performed with this device can be described as an original measurement equation in A . However, relative to the earlier reference unit A' , the same measurement equation already represents a physical converted measurement equation. Thus, whether a measurement equation is regarded as original or physically converted depends on the chosen reference unit.

Consequently, every numerical value admissible in $D_{\text{orig}}^{(M)}$ can also occur in $D_{\text{conv}}^{(M)}$. Hence,

$$D_{\text{orig}}^{(M)} \subseteq D_{\text{conv}}^{(M)}.$$

Since

$$D_{\text{orig}}^{(M)} = D,$$

it follows that

$$D \subseteq D_{\text{conv}}^{(M)}.$$

At the same time, a physical converted measurement equation remains a physical measurement equation and is therefore subject to the same requirement of finite numerical representability. Its admissible numerical values must consequently belong to the set D of finitely representable numerical values. Hence,

$$D_{\text{conv}}^{(M)} \subseteq D.$$

Hence,

$$D \subseteq D_{\text{conv}}^{(M)} \subseteq D,$$

which implies

$$D_{\text{conv}}^{(M)} = D.$$

Together with

$$D_{\text{orig}}^{(M)} = D,$$

we therefore obtain

$$D_{\text{orig}}^{(M)} = D_{\text{conv}}^{(M)} = D.$$

□

Lemma 3.14. *The equality of the admissible numerical sets established for an individual measurement in Lemma 3.13 also holds for the measurement framework as a whole. Thus,*

$$D_{\text{orig}} = D_{\text{conv}} = D.$$

Proof. The argument of Lemma 3.13 does not depend on any particular property of the individual measurement M . It could therefore be applied directly to every admissible measurement to obtain the corresponding equality for the measurement framework as a whole.

The following construction provides an additional direct demonstration of this general result.

Consider an arbitrary original measurement equation expressed in a reference length unit A ,

$$\bar{x}_A = x_A + x_{\text{ba},A}.$$

Choose an arbitrary nonzero integer N . Multiplying the complete equation by N gives

$$N\bar{x}_A = Nx_A + Nx_{\text{ba},A},$$

with length unit C .

Since multiplication by an integer preserves finite representability, all three numerical values remain finite decimals. The resulting equation is therefore again a possible physical measurement equation.

Moreover, by Lemma 3.12, every finite decimal is independently admissible as the numerical representation of $\bar{x}$, x , and x_{ba} . The scaled equation may therefore occur as an original measurement equation of an independent measurement expressed in length unit C .

We chose C such that a unit conversion from C to A by the conversion factor $1/N$ results in a physical converted equation in length unit A with the exact numerical values for $\bar{x}$, x , and x_{ba} as in our considered arbitrary original measurement equation

$$\bar{x}_A = x_A + x_{\text{ba},A}.$$

Since the original measurement equation in A was chosen arbitrarily, every admissible numerical representation occurring in an original measurement equation can also occur in a physical converted measurement equation. Consequently,

$$D_{\text{orig}} \subseteq D_{\text{conv}}.$$

For the converse direction, consider an arbitrary physical converted measurement equation expressed in a reference length unit B .

By Lemma 3.12, every finite decimal is independently admissible as the numerical representation of $\bar{x}$, x , and x_{ba} . Therefore, the numerical values of the physically converted equation in B can be reproduced by an independently chosen measurement with an appropriate measurement configuration.

This measurement can be performed with a measuring device whose native reference unit is B . The resulting equation then has the same numerical representation as the physical converted measurement equation, but is an original measurement equation for this measuring device.

Since the physical converted measurement equation in B was chosen arbitrarily, every admissible numerical representation occurring in a physical converted measurement equation can therefore also occur in an original measurement equation. Consequently,

$$D_{\text{conv}} \subseteq D_{\text{orig}}.$$

The two inclusions imply

$$D_{\text{orig}} = D_{\text{conv}}.$$

By Lemma 3.12, the admissible numerical values of the measurement framework are given by D . Therefore,

$$D_{\text{orig}} = D_{\text{conv}} = D.$$

□

3.8 Concluding Remark on the Representation Framework

The purpose of the preceding construction was not to develop a specific physical model of measurement. It was to identify the arithmetic structure shared by all exact physical position measurements that satisfy the axioms introduced in Section 2.

The measurement framework shows that every exact physical position measurement has a canonical arithmetic representation of the form given in Eq. 3.5. This representation does not depend on the measuring device, the experimental setup, the physical interaction mechanism, or the dynamical theory.

Furthermore, the construction does not require or imply that the underlying ontic position space is discrete. The restriction to finite numerical representations applies only to the numerical encoding of exact measurement results.

The preceding lemmas establish the admissible numerical sets at three distinct levels. Lemma 3.12 shows that every finite decimal is independently admissible as the numerical representation of $\bar{x}$, x , and x_{ba} , so that

$$D_{\bar{x}} = D_x = D_{x_{\text{ba}}} = D.$$

Lemma 3.13 then establishes for an individual measurement M that the admissible numerical set is preserved under a change of the reference length unit,

$$D_{\text{orig}}^{(M)} = D_{\text{conv}}^{(M)} = D.$$

Finally, Lemma 3.14 extends this result to the measurement framework as a whole and establishes the more general result

$$D_{\text{orig}} = D_{\text{conv}} = D.$$

These results establish the numerical representation structure used in the following proof. The contradiction derived below therefore concerns the compatibility of this structure with the arithmetic properties of Eq. 3.5 under the assumptions of the present measurement framework.

4 Main Result

The following result applies to theories satisfying the preceding axioms and admitting at least one exact position measurement with non-vanishing positional back-action, $x_{\text{ba}} \neq 0$.

According to Lemma 3.12, every finitely representable numerical value belonging to D may occur independently as the numerical representation of the final ontic position $\bar{x}$, the initial ontic position x , and the measurement back-action x_{ba} in an exact physical measurement equation. By Lemma 3.13, this admissible numerical set is the same for original and physical converted measurement equations of the individual measurement considered here. Consequently, every finitely representable decimal must be capable of appearing in each of the three quantities individually.

Let

$$\frac{q}{2^o 5^p}$$

be an arbitrary finite decimal satisfying

$$\gcd(q, 2^o 5^p) = 1,$$

where

$$q \in \mathbb{Z} \setminus \{0\}, \quad o, p \in \mathbb{Z}.$$

Since every converted measurement equation originates from an original measurement equation, we use the arithmetic measurement Eq. (3.5) as the original equation and investigate whether the arbitrary finite decimal

$$\frac{q}{2^o 5^p}$$

can become the numerical representation of the final ontic position, the initial ontic position, and the measurement back-action in three different physical converted measurement equations.

Throughout the following derivation, we consider one fixed original measurement equation and derive the resulting arithmetic restrictions for the corresponding physical converted measurement equations.

The purpose of the main proof is therefore to establish the contradiction at the level of an individual measurement construction. The measurement-specific integers a , c , e , i , j , k , l , m , and n are therefore fixed by the measurement under consideration.

At the same time, the derivation must not depend on their particular numerical values. The resulting arithmetic restrictions must therefore hold for every admissible choice of these quantities within the measurement construction; in other words, the subsequent argument must be independent of the measurement-specific integers.

We first require

$$\frac{q}{2^o 5^p}$$

to become the numerical representation of the final ontic position.

Dividing Eq. (3.5) by

$$\bar{x} = \frac{a}{2^i 5^j}$$

and multiplying the resulting equation by

$$\frac{q}{2^o 5^p}$$

corresponds to a unit conversion with conversion factor

$$\frac{q}{2^o 5^p} \cdot \frac{2^i 5^j}{a}.$$

The quantities in the converted measurement equation are indexed by c_1 .

This yields

$$\underbrace{\frac{q}{2^o 5^p}}_{\bar{x}_{(c_1)}} = \underbrace{\frac{qc}{a 2^{k-i+o} 5^{l-j+p}}}_{x_{(c_1)}} + \underbrace{\frac{qe}{a 2^{m-i+o} 5^{n-j+p}}}_{x_{\text{ba}(c_1)}}. \quad (4.1)$$

By construction, the final ontic position already possesses the desired finite representation. For Eq. (4.1) to remain a physical converted measurement equation, both remaining quantities must likewise possess finite numerical representations.

Lemma 3.10 established that

$$\gcd(a, c) = 1$$

and

$$\gcd(a, e) = 1.$$

Hence, the only possible obstruction to finite representability of the two remaining quantities is the factor a occurring in the denominator. Therefore Eq. (4.1) remains physical if and only if

$$\frac{q}{a} \in \mathbb{Z} \setminus \{0\}. \quad (4.2)$$

Equation (4.2) therefore provides the necessary and sufficient arithmetic condition under which the arbitrary finite decimal

$$\frac{q}{2^o 5^p}$$

can occur as the numerical representation of a final ontic position in a physical converted measurement equation.

An important point already appears at this stage. Unlike a purely mathematical scaling, a physical unit conversion is not always admissible. It must satisfy an arithmetic consistency condition. If condition (4.2) is violated, the converted equation still exists as a mathematical

identity, but it no longer represents a physical measurement equation because one or more quantities no longer have finite numerical representations.

Condition (4.2) already shows the first part of the contradiction. For the converted measurement equation to remain physical, the admissible numerical representations are no longer arbitrary. They are restricted by the measurement-specific parameter a . Therefore, the admissible numerical set for the converted equation is no longer guaranteed to satisfy

$$D_{\text{conv}}^{(M)} = D,$$

as required by Lemma 3.13.

At this stage, however, the restriction still depends on the measurement-specific parameter a . It therefore does not yet establish a contradiction for the individual measurement that is independent of any measurement-specific parameter. To determine the full arithmetic restriction imposed on the admissible numerical values of this measurement, the same analysis must also be applied to the other two quantities x and x_{ba} .

We now require the same arbitrary finite decimal

$$\frac{q}{2^o 5^p}$$

to become the numerical representation of the initial ontic position. The procedure is completely analogous. We divide Eq. (3.5) by

$$x = \frac{c}{2^k 5^l}$$

and multiply by

$$\frac{q}{2^o 5^p}.$$

The converted quantities are indexed by c_2 , yielding

$$\underbrace{\frac{qa}{c 2^{i-k+o} 5^{j-l+p}}}_{\bar{x}_{(c_2)}} = \underbrace{\frac{q}{2^o 5^p}}_{x_{(c_2)}} + \underbrace{\frac{qe}{c 2^{m-k+o} 5^{n-l+p}}}_{x_{\text{ba}(c_2)}}. \quad (4.3)$$

Again, the initial ontic position already possesses the desired finite numerical representation by construction. The converted equation remains physical only if both remaining quantities continue to admit finite decimal representations.

From Lemma 3.10 we have

$$\gcd(a, c) = 1$$

and

$$\gcd(c, e) = 1.$$

Therefore the only possible obstruction is the denominator c . Consequently, Eq. (4.3) remains physical if and only if

$$\frac{q}{c} \in \mathbb{Z} \setminus \{0\}. \quad (4.4)$$

Equation (4.4) is therefore the necessary and sufficient arithmetic condition under which the arbitrary finite decimal

$$\frac{q}{2^o 5^p}$$

can occur as the numerical representation of an initial ontic position in a physical converted measurement equation.

Lemma 3.10 guarantees that a and c are coprime,

$$\gcd(a, c) = 1,$$

while Condition 3.9 requires their absolute values to be distinct,

$$|a| \neq |c|.$$

Thus, combining Eqs. (4.2) and (4.4) gives

$$\frac{q}{ac} \in \mathbb{Z} \setminus \{0\}.$$

Since a and c are coprime and $|a| \neq |c|$, they cannot both have absolute value one. Hence $|ac| > 1$. It follows that ac cannot divide $q = \pm 1$, and therefore

$$q \neq \pm 1,$$

which is the measurement-independent restriction that we were looking for. However, this result is not yet fully symmetric with respect to the three quantities $\bar{x}$, x , and x_{ba} . The remaining conversion is therefore required to establish the complete structural restriction without privileging any particular term of the measurement equation.

Hence, we finally require the same arbitrary finite decimal

$$\frac{q}{2^o 5^p}$$

to become the numerical representation of the measurement back-action. Dividing Eq. (3.5) by

$$x_{\text{ba}} = \frac{e}{2^m 5^n}$$

and multiplying by

$$\frac{q}{2^o 5^p}$$

produces a third converted measurement equation, whose quantities are indexed by c_3 :

$$\underbrace{\frac{aq}{e 2^{i-m+o} 5^{j-n+p}}}_{\bar{x}(c_3)} = \underbrace{\frac{cq}{e 2^{k-m+o} 5^{l-n+p}}}_{x(c_3)} + \underbrace{\frac{q}{2^o 5^p}}_{x_{\text{ba}}(c_3)}. \quad (4.5)$$

Since the measurement back-action already possesses the desired finite representation, physical admissibility requires only that the remaining two quantities remain finitely representable.

Applying Lemma 3.10 once more,

$$\gcd(a, e) = 1$$

and

$$\gcd(c, e) = 1,$$

shows that this is equivalent to the condition

$$\frac{q}{e} \in \mathbb{Z} \setminus \{0\}. \quad (4.6)$$

Equation (4.6) is therefore the necessary and sufficient arithmetic condition under which the arbitrary finite decimal

$$\frac{q}{2^o 5^p}$$

can occur as the numerical representation of the measurement back-action in a physical converted measurement equation.

Including the third conversion completes the symmetry of the argument. At this point, the three conversion procedures have established the complete set of arithmetic conditions that every arbitrary finite decimal must satisfy in order to remain physically admissible. Since, according to Lemma 3.12, every finitely representable numerical value must be capable of occurring independently as a final ontic position, an initial ontic position, and a measurement back-action, conditions (4.2), (4.4), and (4.6) must therefore hold simultaneously for every admissible finite decimal.

Since the integers a , c , and e are pairwise coprime by Lemma 3.10, conditions (4.2), (4.4), and (4.6) can be combined into a single arithmetic condition,

$$\frac{q}{ace} \in \mathbb{Z} \setminus \{0\}. \quad (4.7)$$

Condition (4.7) appears to depend on the particular original measurement through the measurement-specific integers a , c , and e , but this dependence disappears once again by taking into account Lemma 3.10, which guarantees that a , c , and e are pairwise coprime,

$$\gcd(a, c) = \gcd(a, e) = \gcd(c, e) = 1,$$

and Condition 3.9, which requires that their absolute values are pairwise distinct,

$$|a| \neq |c|, \quad |a| \neq |e|, \quad |c| \neq |e|.$$

Consequently, condition (4.7) requires the numerator q to possess three pairwise coprime divisors whose absolute values are mutually distinct. Any integer that does not admit such a divisor structure is excluded. Positive and negative prime powers, as well as ± 1 , are immediate examples that will be analyzed below.

We first examine the possibility $q = \pm 1$. Since every divisor of ± 1 has absolute value one, condition (4.7) would imply

$$|a| = |c| = |e| = 1,$$

which immediately violates Condition 3.9. Therefore,

$$q \neq \pm 1. \quad (4.8)$$

Second, suppose q is a positive or negative prime power,

$$q = \pm d^f,$$

where d is a prime number ($d \in \mathbb{P}$) and f is a positive integer.

Every non-trivial divisor of a prime power is itself a power of the same prime d . Consequently, any two non-trivial divisors necessarily possess the common prime factor d and therefore cannot be coprime. Condition (4.7), however, requires the existence of three pairwise coprime divisors a , c , and e . Hence, at most one of these divisors can be non-trivial. The remaining divisors must therefore have absolute value one. This is impossible because Condition 3.9 simultaneously requires the absolute values of a , c , and e to be pairwise distinct. Accordingly, q cannot be a positive or negative prime power.

We therefore conclude that

$$q \neq \pm d^f \quad (4.9)$$

for every prime number d and every positive integer f .

The restrictions Eqs. (4.8) and (4.9) already arise in the idealized localization limit. Their origin cannot be attributed to finite detector resolution, uncertainty relations, imperfect measuring devices, or any other physical approximation. They follow exclusively from the assumption that ontic positions admit a consistent numerical representation together with the arithmetic structure established by Eq. 3.5, Lemma 3.10, Lemma 3.12, and Condition 3.9.

More importantly, the restrictions in Eqs. 4.8 and 4.9 are completely independent of the particular measurement-specific values a , c , e , i , j , k , l , m , and n . Thus, within the measurement construction considered here, they apply to every admissible choice of these quantities.

An important consequence follows immediately for the individual measurement M considered in the main proof. Every finite decimal whose numerator satisfies

$$q = \pm 1$$

or

$$q = \pm d^f$$

is excluded from the set of admissible numerical representations of physical converted measurement equations. Consequently, for the individual measurement M considered here, the admissible set of numerical values for physical converted measurement equations is strictly smaller than the admissible set for original measurement equations. However, Lemma 3.13 established that original and physical converted measurement equations possess exactly the same set of admissible numerical representations. In particular,

$$D_{\text{orig}}^{(M)} = D_{\text{conv}}^{(M)}. \quad (4.10)$$

Equation (4.10) expresses the invariance of admissible numerical representations under all physical unit conversions. Every finite decimal that can occur in an original physical

measurement equation must also occur in a physical converted measurement equation, and conversely.

The arithmetic restrictions derived above contradict this result directly. Eqs. (4.8)–(4.9) exclude infinitely many finite decimals from $D_{\text{conv}}^{(M)}$, namely all finite decimals whose numerators are equal to ± 1 or to a positive or negative prime power. Consequently,

$$D_{\text{orig}}^{(M)} \supsetneq D_{\text{conv}}^{(M)}. \quad (4.11)$$

Eqs. (4.10) and (4.11) are mutually incompatible.

The contradiction is therefore obtained already for the individual measurement construction considered in the main proof.

The contradiction cannot be attributed to finite measurement precision, detector resolution, uncertainty-related limitations, or to the particular numerical values of the measurement-specific quantities.

The main proof derives all three arithmetic conditions from a single original measurement equation. One may therefore ask whether the contradiction depends on this common origin.

Appendix A shows that this is not the case. There, each of the three arithmetic conditions is derived from a different and independently chosen original measurement equation. The same contradiction is therefore obtained without assuming a common original measurement equation.

In this general case, the admissible numerical sets are no longer associated with a particular measurement M , but with the measurement framework as a whole. Lemma 3.14 requires for this general framework that

$$D_{\text{orig}} = D_{\text{conv}} = D.$$

Appendix A, however, shows that the independent measurements lead to

$$D_{\text{orig}} \supsetneq D_{\text{conv}}.$$

The contradiction therefore persists even when the three arithmetic conditions do not originate from a common original measurement equation.

Hence, the Ontic Position Hypothesis cannot be consistently maintained under the stated axioms and the introduced measurement framework.

The argument does not depend on the particular numerical values of the measurement-specific quantities a , c , and e . These quantities enter only as arbitrary numerical representations of the respective physical quantities, and their particular values are not required at any stage of the derivation. The resulting contradiction is therefore independent of the particular measurement values and, in this sense, measurement independent.

A further objection concerns the distinction between an ontically defined position and its experimental determination. One may argue that the position entering the measurement equation cannot be determined exactly because of measurement uncertainty, including the Heisenberg uncertainty relation. As shown in Appendix B, such uncertainty can be represented by an additional unknown positional contribution. The same appendix also shows that coordinate perturbations and spatial extension do not remove the contradiction as long as the fundamental entity retains an ontically defined spatial property admitting a numerical representation.

5 Conclusion

5.1 Physical Implications and Scope of the No-Go Theorem

The contradiction is established at two levels. For the individual measurement construction considered in the main proof, Lemma 3.13 requires

$$D_{\text{orig}}^{(M)} = D_{\text{conv}}^{(M)} = D,$$

whereas the arithmetic analysis yields

$$D_{\text{orig}}^{(M)} \supsetneq D_{\text{conv}}^{(M)}.$$

Appendix A removes the requirement of a common original measurement equation by obtaining the three required numerical constraints from independently chosen original measurement equations. For this general measurement framework, Lemma 3.14 requires

$$D_{\text{orig}} = D_{\text{conv}} = D,$$

whereas Appendix A yields

$$D_{\text{orig}} \supsetneq D_{\text{conv}}.$$

The contradiction therefore holds both for the individual measurement construction and for the general measurement framework. It does not depend on the three arithmetic constraints originating from a common original measurement equation.

A specific limiting case remains outside the scope of the present no-go result. If a theory permits only exact position measurements satisfying

$$x_{\text{ba}}^{(M)} = 0$$

for every admissible measurement M , the three-term arithmetic structure used in the proof does not arise. The existence of some measurements with vanishing positional back-action is not sufficient to evade the result: as long as at least one exact position measurement with $x_{\text{ba}}^{(M)} \neq 0$ is admitted, the measurement class considered in the proof remains available.

The theorem is further restricted to ontic positions admitting an injective numerical representation. Formally, fundamentally non-representable ontic positions therefore remain outside its scope. However, assigning a position to a physical entity is conventionally associated with a consistent coordinate representation. Once such a coordinate assignment exists, an injective numerical representation follows naturally. Avoiding the theorem therefore requires either the special case of universally vanishing positional back-action described above or a notion of fundamental position that does not satisfy the representation assumptions of the present framework.

The remaining ontological possibilities may consequently involve more abstract structures, for example relational, algebraic, information-theoretic, or emergent frameworks [3–7]. Algebraic approaches to quantum field theory provide an important example of a formalism in which the theoretical structure is formulated in terms of local algebras of observables rather than a fundamental ontology of localized particles [8, 9].

5.2 Interpretation of the Result

The theorem does not merely constrain a particular class of hidden-variable theories or ontological particle models [10]. Under the assumptions of the present framework, it constrains fundamental physical entities possessing ontic positions in the conventional sense of uniquely assignable numerical coordinate locations.

The result also applies to fundamentally classical particle ontologies whenever they satisfy these assumptions. In particular, a classical theory in which particles possess exact, numerically representable ontic positions and which admits at least one exact position measurement with non-vanishing positional back-action falls within the scope of the no-go theorem. This does not affect the validity of classical mechanics as an effective description within its established domain of applicability.

The same observation extends to physical quantities defined through exact spatial localization, such as geometrical lengths or characteristic spatial scales, whenever they depend on fundamentally well-defined positional assignments.

Importantly, the theorem does not exclude emergent or effective notions of position. It constrains only fundamental ontic positions satisfying the assumptions of the present framework. It is therefore compatible with approaches in which position arises only at an effective or emergent level.

This conclusion should not be understood as selecting a particular alternative ontology. The ontology of quantum field theory itself remains nontrivial: both particle and field interpretations face independent conceptual difficulties, and the existence of field quanta or a Fock-space representation does not by itself establish a fundamental particle ontology [11, 12].

The theorem consequently places a constraint on the ontology of a fundamental particle without determining which alternative ontology, if any, provides the correct fundamental description.

Statements and Declarations

Funding

The author received no funding for this work.

Competing Interests

The author declares no competing interests.

Data Availability

No datasets were generated or analyzed during the current study.

Author Contributions

The author was solely responsible for the conceptualization, formal analysis, investigation, methodology, and writing of this work.

A Independence of the Original Measurement Equation

The main proof derives the contradiction at the level of an individual measurement construction. It starts from one original measurement equation and generates three physical converted measurement equations from it.

Lemma 3.14 establishes, however, that the admissible numerical set is universal within the measurement framework and does not depend on the particular original measurement equation.

The present appendix considers the more general case of three independent original physical measurement equations. It demonstrates explicitly that the same arithmetic obstruction and the same contradiction are obtained when the three relevant constraints originate from independent measurements.

Suppose that three independent original measurement equations are given by

$$\frac{a_1}{2^{i_1}5^{j_1}} = \frac{c_1}{2^{k_1}5^{l_1}} + \frac{e_1}{2^{m_1}5^{n_1}}. \quad (\text{A.1})$$

$$\frac{a_2}{2^{i_2}5^{j_2}} = \frac{c_2}{2^{k_2}5^{l_2}} + \frac{e_2}{2^{m_2}5^{n_2}}. \quad (\text{A.2})$$

$$\frac{a_3}{2^{i_3}5^{j_3}} = \frac{c_3}{2^{k_3}5^{l_3}} + \frac{e_3}{2^{m_3}5^{n_3}}. \quad (\text{A.3})$$

where each equation possesses its own measurement-specific integers indicated by the subscripts 1, 2, and 3, respectively.

Each original measurement equation satisfies the assumptions established in the previous sections. Consequently, Lemma 3.10 applies independently to every measurement equation,

$$\gcd(a_i, c_i) = \gcd(a_i, e_i) = \gcd(c_i, e_i) = 1, \quad i = 1, 2, 3. \quad (\text{A.4})$$

Furthermore, Condition 3.9 requires

$$|a_i| \neq |c_i|, \quad |a_i| \neq |e_i|, \quad |c_i| \neq |e_i|. \quad (\text{A.5})$$

Suppose now that the same arbitrary finite decimal

$$\frac{q}{2^o5^p}$$

is required to occur as a final ontic position in the first converted equation, an initial ontic position in the second converted equation, and a measurement back-action in the third converted equation.

Repeating the derivation of the main proof for the three independent original equations yields

$$\frac{q}{a_1} \in \mathbb{Z} \setminus \{0\}, \quad (\text{A.6})$$

$$\frac{q}{c_2} \in \mathbb{Z} \setminus \{0\}, \quad (\text{A.7})$$

$$\frac{q}{e_3} \in \mathbb{Z} \setminus \{0\}. \quad (\text{A.8})$$

The three integers a_1 , c_2 , and e_3 need not initially be pairwise coprime since they originate from three independent measurements. If this is the case, the coordinate origins can be translated by a suitably chosen finite displacement to change the numerical values of a_1 and c_2 . This is the same logic that underlies Condition 3.9. Under a translation of the coordinate origin, the measurement back-action x_{ba} , and hence e_3 , remains unchanged, since it is simply the fixed positional difference between the final and initial positions, $x_{\text{ba}} = \bar{x} - x$. Such a finite displacement preserves the physicality of the measurement since all three quantities $\bar{x}$, x , and x_{ba} remain finite decimals in their respective equations.

Example (Coordinate Translation). Consider a physical measurement equation of the form

$$\bar{x} = x + x_{\text{ba}}.$$

Displacing the coordinate origin by the finite decimal x_{disp} leaves the measurement back-action x_{ba} unchanged, since $x_{\text{ba}} = \bar{x} - x$, whereas the same coordinate displacement is added to both the initial and final positions:

$$\underbrace{(\bar{x} + x_{\text{disp}})}_{\bar{x}_{\text{new}}} = \underbrace{(x + x_{\text{disp}})}_{x_{\text{new}}} + x_{\text{ba}}.$$

This equation remains physical. Only the numerical values assigned to the initial and final positions have changed. Hence, a finite displacement of a coordinate origin can modify the relevant numerical values a_1 and c_2 accordingly. Therefore, without loss of generality, the integers a_1 , c_2 , and e_3 can always be chosen such that

$$\gcd(a_1, c_2) = \gcd(a_1, e_3) = \gcd(c_2, e_3) = 1.$$

Now the three conditions can be combined into

$$\frac{q}{a_1 c_2 e_3} \in \mathbb{Z} \setminus \{0\}. \quad (\text{A.9})$$

From here, the argument of the main theorem applies without modification.

The contradiction with the universal admissible set established by Lemma 3.14 therefore follows in exactly the same way as in the main proof.

In particular, the independent-measurement construction yields

$$D_{\text{orig}} \supsetneq D_{\text{conv}},$$

while Lemma 3.14 requires

$$D_{\text{orig}} = D_{\text{conv}} = D.$$

Thus, the contradiction is not an artifact of using a single original measurement equation. It persists when the three relevant constraints are obtained from three independent original physical measurements.

The no-go theorem is therefore independent of the particular choice of the original measurement equation and of whether the three constraints are taken from one measurement construction or from independent measurements.

B Independence from Measurement and Ontological Modifications

The main result does not depend on the numerical knowledge of the measurement-specific quantities appearing in the construction. Nevertheless, one may object that the positional quantities entering the measurement equation are not exactly accessible because of measurement uncertainties, including uncertainties associated with the Heisenberg uncertainty relation. This appendix shows that introducing such uncertainties does not alter the arithmetic structure responsible for the contradiction.

The analysis in this appendix is formulated for the individual measurement construction considered in the main proof and therefore corresponds to Lemma 3.13.

The more general case follows in the same way from the independent-measurement construction of Appendix A. The additional positional contributions introduced below do not change the arithmetic conditions obtained from the individual conversions. Therefore, the same argument can be applied separately to each of the three independent measurements considered in Appendix A. Consequently, the result also applies to the general measurement framework of Lemma 3.14.

B.1 Positional Uncertainty and the Heisenberg Relation

Consider an unknown positional contribution Δx between the underlying ontic position and the numerical quantities entering the measurement equation. The physical measurement equation can then be written as

$$\bar{x} = x + x_{\text{ba}} + \Delta x. \quad (\text{B.1})$$

Here, Δx denotes a general unknown positional contribution. It may, for example, represent an uncertainty associated with the measurement process, including an uncertainty motivated by the Heisenberg uncertainty relation, or any other unknown positional fluctuation or perturbation. The argument below does not depend on the physical origin or numerical knowledge of Δx .

Since $\bar{x}$, x , and x_{ba} belong to the admissible numerical set $D_{\text{orig}}^{(M)}$, Eq. (B.1) implies that Δx must likewise belong to $D_{\text{orig}}^{(M)}$. It can therefore be written in the form

$$\Delta x = \frac{g}{2^r 5^s}, \quad \gcd(g, 2^r 5^s) = 1, \quad g \in \mathbb{Z} \setminus \{0\}, \quad r, s \in \mathbb{Z}. \quad (\text{B.2})$$

This requirement does not imply that the value of Δx is known. It only follows from the requirement that the physical measurement equation remains within the admissible numerical set $D_{\text{orig}}^{(M)}$.

The introduction of Δx does not modify the conversion argument of the main theorem. The same arbitrary finite numerical value

$$\frac{q}{2^o 5^p}, \quad (\text{B.3})$$

may be considered as the target numerical value in each of the three coordinate-conversion procedures used in the main theorem.

In the first conversion, the numerical values of $\bar{x}$, x , and x_{ba} are transformed by the same coordinate-conversion factor as in the original argument. The additional contribution Δx is

transformed by the same factor. Its numerator therefore acquires the additional integer factor g , but no new arithmetic condition involving g is required. The condition obtained from the first conversion remains

$$\frac{q}{a} \in \mathbb{Z} \setminus \{0\}. \quad (\text{B.4})$$

Indeed, the corresponding transformed contribution from Δx contains the factor

$$\frac{qg}{a} = \left(\frac{q}{a}\right)g, \quad (\text{B.5})$$

which is an integer whenever Eq. (B.4) holds. No coprimality condition between g and a is required.

The same reasoning applies to the second and third conversions. The corresponding conditions remain

$$\frac{q}{c} \in \mathbb{Z} \setminus \{0\}, \quad (\text{B.6})$$

and

$$\frac{q}{e} \in \mathbb{Z} \setminus \{0\}. \quad (\text{B.7})$$

The additional factor g does not modify either condition, since

$$\frac{qg}{c} = \left(\frac{q}{c}\right)g$$

and

$$\frac{qg}{e} = \left(\frac{q}{e}\right)g$$

are integers whenever Eqs. (B.6) and (B.7) hold, respectively.

Consequently, the three arithmetic conditions obtained in the main theorem remain unchanged. Since a , c , and e are pairwise coprime by Lemma 3.10, they combine to give

$$\frac{q}{ace} \in \mathbb{Z} \setminus \{0\}. \quad (\text{B.8})$$

Thus, the additional positional contribution Δx does not alter the arithmetic condition responsible for the contradiction.

Moreover, no additional condition on the numerical set to which the transformed contribution Δx belongs is required. The derivation of the contradiction depends only on the requirement that the three quantities $\bar{x}$, x , and x_{ba} remain in the admissible numerical set $D_{\text{conv}}^{(M)}$, as guaranteed by Lemma 3.13. The transformed contribution Δx is therefore irrelevant to the arithmetic conditions generating the contradiction.

Importantly, this conclusion does not depend on numerical knowledge of a , c , e , or g . The argument requires only that the corresponding physical quantities belong to the admissible numerical set $D_{\text{orig}}^{(M)}$. Consequently, introducing an unknown positional contribution cannot restore consistency. It imposes an additional requirement—namely, that Δx itself belong to $D_{\text{orig}}^{(M)}$ in the original physical measurement equation—but does not remove any of the requirements already responsible for the contradiction.

The Heisenberg uncertainty relation therefore does not provide an independent escape from the theorem. The uncertainty relation concerns the limitations on the simultaneous determination of physical quantities such as position and momentum; it does not by itself

establish the non-existence of an optically defined position. If an optically defined position is retained, but its value is experimentally inaccessible with arbitrary precision, the unknown contribution associated with this inaccessibility can be represented by Δx without changing the arithmetic structure of the proof.

Conversely, if a theory rejects the existence of an optically defined position altogether, it does not constitute a counterexample to the theorem, but instead lies outside the class of theories to which the theorem applies. The distinction between optic existence and experimental determinability is therefore essential.

Thus, the introduction of an unknown positional contribution does not remove the contradiction for the individual measurement construction considered in the main proof and described by Lemma 3.13.

By applying the same argument independently to the three original physical measurements of Appendix A, the same conclusion follows for the general measurement framework of Lemma 3.14.

B.2 Perturbations of the Coordinate System

A related objection is that the coordinate system itself may be subject to unknown perturbations, such that the numerical values appearing in the measurement equation represent only perturbed or approximate coordinates. Suppose that independent perturbations are associated with the three quantities of the original measurement equation. The resulting relation may be written as

$$\bar{x} + \Delta x_1 = x + \Delta x_2 + x_{ba} + \Delta x_3. \quad (\text{B.9})$$

All perturbations in Eq. (B.9) enter additively. They can therefore be combined into a single effective contribution,

$$\Delta x = \Delta x_1 + \Delta x_2 + \Delta x_3. \quad (\text{B.10})$$

Equation (B.9) can consequently be rewritten in the form

$$\bar{x} = x + x_{ba} + \Delta x, \quad (\text{B.11})$$

which is identical in structure to Eq. (B.1).

The three individual perturbations therefore introduce no new mathematical structure into the argument. Since they enter only through their additive combination, their net contribution can be represented by a single effective positional term Δx . The requirement for Δx is analogous to equation (B.2). Thus, the effective contribution Δx yields a finite numerical representation within the extended measurement equation (B.11). Once this effective contribution has been introduced, the analysis of the preceding subsection applies directly. In particular, the three coordinate-conversion procedures remain subject to the conditions

$$\frac{q}{a} \in \mathbb{Z} \setminus \{0\}, \quad \frac{q}{c} \in \mathbb{Z} \setminus \{0\}, \quad \frac{q}{e} \in \mathbb{Z} \setminus \{0\}, \quad (\text{B.12})$$

and hence to

$$\frac{q}{ace} \in \mathbb{Z} \setminus \{0\}. \quad (\text{B.13})$$

Thus, perturbations of the coordinate system, or multiple additive perturbations of the spatial description, can be absorbed into a single effective positional contribution. They therefore do not provide a separate mechanism for avoiding the contradiction established by the main theorem.

B.3 Spatially Extended Entities and Field Excitations

A further possibility is to abandon the point-particle description and regard the fundamental entity instead as a spatially extended wave or as an excitation of a field. Such an entity is not pointlike and might therefore initially appear to avoid the no-go result.

The relevant issue, however, is not whether the fundamental entity is pointlike or spatially extended, but whether it possesses an optically defined spatial property that admits a consistent numerical representation. For example, a uniquely defined spatial feature of a field configuration $\phi(x, t)$, such as a maximum, may be assigned a position

$$x_{\max} = \arg \max_x \phi(x, t). \quad (\text{B.14})$$

If $x_{\max}$ is regarded as an optically defined property of the fundamental entity and is numerically represented in the same position space as the positions considered in the theorem, its determination is subject to the same numerical representation and measurement structure as any other optically defined position. In particular, the corresponding physical measurement must satisfy the same measurement equation and the same requirements on admissible numerical representations. The fact that the spatial property belongs to an extended field configuration rather than to a pointlike entity does not alter this structure.

Consequently, replacing a point particle by a spatially extended wave or field excitation does not, by itself, provide an escape from the no-go result. An escape would require the fundamental entity to lack any optically defined spatial property to which the numerical representation assumed by the theorem applies.

More generally, stochastic or deterministic modifications, as well as an increase in the number of spatial dimensions, do not by themselves alter the conclusion. As long as a fundamental entity is assumed to possess an optically defined and numerically representable position in space, the no-go result applies.

The result therefore does not exclude classical objects or spatially extended physical structures as such. It excludes only those fundamental descriptions in which such an entity is assigned an optically defined and numerically representable position satisfying the assumptions of the theorem.

References

- [1] David B. Malament. In defense of dogma: Why there cannot be a relativistic quantum mechanics of (localizable) particles. In Rob Clifton, editor, *Perspectives on Quantum Reality*, pages 1–10. Kluwer Academic Publishers, 1996. doi: 10.1007/978-94-015-8656-6_1.
- [2] Hans Halvorson and Rob Clifton. No place for particles in relativistic quantum theories? *Philosophy of Science*, 69(1):1–28, 2002. doi: 10.1086/338939.
- [3] Carlo Rovelli. Relational quantum mechanics. *International Journal of Theoretical Physics*, 35(8):1637–1678, 1996. doi: 10.1007/BF02302261.
- [4] John Archibald Wheeler. Information, physics, quantum: The search for links. In Wojciech H. Zurek, editor, *Complexity, Entropy and the Physics of Information*, pages 3–28. Addison-Wesley, 1990.

- [5] Mark Van Raamsdonk. Building up spacetime with quantum entanglement. *General Relativity and Gravitation*, 42(10):2323–2329, 2010. doi: 10.1007/s10714-010-1034-0.
- [6] Steven French. *The Structure of the World: Metaphysics and Representation*. Oxford University Press, 2014. doi: 10.1093/acprof:oso/9780199684847.001.0001.
- [7] James Ladyman. What is structural realism? *Studies in History and Philosophy of Science Part A*, 29(3):409–424, 1998. doi: 10.1016/S0039-3681(98)80129-5.
- [8] Rudolf Haag and Daniel Kastler. An algebraic approach to quantum field theory. *Journal of Mathematical Physics*, 5(7):848–861, 1964. doi: 10.1063/1.1704187.
- [9] Rudolf Haag. *Local Quantum Physics: Fields, Particles, Algebras*. Springer, 2 edition, 1996. doi: 10.1007/978-3-642-61458-3.
- [10] Dustin Lazarovici, Andrea Oldofredi, and Michael Esfeld. Observables and unobservables in quantum mechanics: How the no-hidden-variables theorems support the bohmian particle ontology. *Entropy*, 20(5):381, 2018. doi: 10.3390/e20050381.
- [11] David John Baker. Against field interpretations of quantum field theory. *The British Journal for the Philosophy of Science*, 60(3):585–609, 2009. doi: 10.1093/bjps/axp027.
- [12] Tomasz Bigaj. Are field quanta real objects? some remarks on the ontology of quantum field theory. *Studies in History and Philosophy of Modern Physics*, 62:145–157, 2018. doi: 10.1016/j.shpsb.2017.08.001.